11.6 Mecanum drive kinematics

A mecanum drive has four wheels, one on each corner of a rectangular chassis. The wheels have rollers offset at 45 degrees (whether it's clockwise or not varies per wheel). The forces they generate when moving forward are shown in figure 11.3.

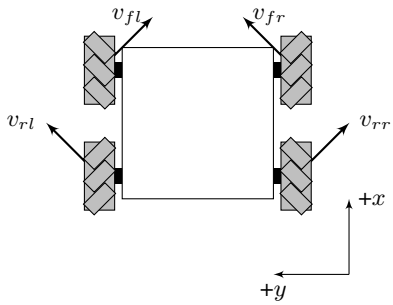

Figure 11.3: Mecanum drive free body diagram

Note that the velocity of the wheel is the same as the velocity in the diagram for the purposes of feedback control. The rollers on the wheel redirect the velocity vector.

11.6.1 Inverse kinematics

First, we'll derive the front-left wheel kinematics. động học bánh trước trái.

vtx vty wxry

$$\vec{v}_{fl} = v_x\hat{i} + v_y\hat{j} + \omega\hat{k} \times (r_{fl_x}\hat{i} + r_{fl_y}\hat{j})$$

$$(=) \quad \vec{v}_{fl} = v_x\hat{i} + v_y\hat{j} + \omega r_{fl_x}\hat{j} - \omega r_{fl_y}\hat{i}$$

$$(=) \quad \vec{v}_{fl} = (v_x - \omega r_{fl_y})\hat{i} + (v_y + \omega r_{fl_x})\hat{j}$$

Project the front-left wheel onto its wheel vector.

$$v_{fl} = ((v_x - \omega r_{fl_y})\hat{i} + (v_y + \omega r_{fl_x})\hat{j}) \cdot \frac{\hat{i} - \hat{j}}{\sqrt{2}}$$

$$v_{fl} = ((v_x - \omega r_{fl_y}) - (v_y + \omega r_{fl_x}))\frac{1}{\sqrt{2}}$$

$$v_{fl} = (v_x - \omega r_{fl_y} - v_y - \omega r_{fl_x})\frac{1}{\sqrt{2}}$$

$$v_{fl} = (v_x - v_y - \omega r_{fl_y} - \omega r_{fl_x})\frac{1}{\sqrt{2}}$$

$$
\begin{aligned}
v_{fl} &= (v_x - v_y - \omega(r_{fl_x} + r_{fl_y}))\frac{1}{\sqrt{2}} \\
v_{fl} &= \frac{1}{\sqrt{2}}v_x - \frac{1}{\sqrt{2}}v_y - \frac{1}{\sqrt{2}}\omega(r_{fl_x} + r_{fl_y})
\end{aligned} \tag{11.8}
$$

Next, we'll derive the front-right wheel kinematics.

$$
\begin{aligned}
\vec{v}_{fr} &= v_x\hat{\imath} + v_y\hat{\jmath} + \omega\hat{k} \times (r_{fr_x}\hat{\imath} + r_{fr_y}\hat{\jmath}) \\
\vec{v}_{fr} &= v_x\hat{\imath} + v_y\hat{\jmath} + \omega r_{fr_x}\hat{\jmath} - \omega r_{fr_y}\hat{\imath} \\
\vec{v}_{fr} &= (v_x - \omega r_{fr_y})\hat{\imath} + (v_y + \omega r_{fr_x})\hat{\jmath}
\end{aligned}
$$

Project the front-right wheel onto its wheel vector.

$$
\begin{aligned}
v_{fr} &= ((v_x - \omega r_{fr_y})\hat{\imath} + (v_y + \omega r_{fr_x})\hat{\jmath}) \cdot (\hat{\imath} + \hat{\jmath})\frac{1}{\sqrt{2}} \\
v_{fr} &= ((v_x - \omega r_{fr_y}) + (v_y + \omega r_{fr_x}))\frac{1}{\sqrt{2}} \\
v_{fr} &= (v_x - \omega r_{fr_y} + v_y + \omega r_{fr_x})\frac{1}{\sqrt{2}} \\
v_{fr} &= (v_x + v_y - \omega r_{fr_y} + \omega r_{fr_x})\frac{1}{\sqrt{2}} \\
v_{fr} &= (v_x + v_y + \omega(r_{fr_x} - r_{fr_y}))\frac{1}{\sqrt{2}} \\
v_{fr} &= \frac{1}{\sqrt{2}}v_x + \frac{1}{\sqrt{2}}v_y + \frac{1}{\sqrt{2}}\omega(r_{fr_x} - r_{fr_y})
\end{aligned} \tag{11.9}
$$

Next, we'll derive the rear-left wheel kinematics.

$$
\begin{aligned}
\vec{v}_{rl} &= v_x\hat{\imath} + v_y\hat{\jmath} + \omega\hat{k} \times (r_{rl_x}\hat{\imath} + r_{rl_y}\hat{\jmath}) \\
\vec{v}_{rl} &= v_x\hat{\imath} + v_y\hat{\jmath} + \omega r_{rl_x}\hat{\jmath} - \omega r_{rl_y}\hat{\imath} \\
\vec{v}_{rl} &= (v_x - \omega r_{rl_y})\hat{\imath} + (v_y + \omega r_{rl_x})\hat{\jmath}
\end{aligned}
$$

Project the rear-left wheel onto its wheel vector.

$$
\begin{aligned}
v_{rl} &= ((v_x - \omega r_{rl_y})\hat{\imath} + (v_y + \omega r_{rl_x})\hat{\jmath}) \cdot (\hat{\imath} + \hat{\jmath})\frac{1}{\sqrt{2}} \\
v_{rl} &= ((v_x - \omega r_{rl_y}) + (v_y + \omega r_{rl_x}))\frac{1}{\sqrt{2}}
\end{aligned}
$$

$$
\begin{aligned}
v_{rl} &= (v_x - \omega r_{rl_y} + v_y + \omega r_{rl_x})\frac{1}{\sqrt{2}} \\
v_{rl} &= (v_x + v_y - \omega r_{rl_y} + \omega r_{rl_x})\frac{1}{\sqrt{2}} \\
v_{rl} &= (v_x + v_y + \omega(r_{rl_x} - r_{rl_y}))\frac{1}{\sqrt{2}} \\
v_{rl} &= \frac{1}{\sqrt{2}}v_x + \frac{1}{\sqrt{2}}v_y + \frac{1}{\sqrt{2}}\omega(r_{rl_x} - r_{rl_y})
\end{aligned} \tag{11.10}
$$

Next, we'll derive the rear-right wheel kinematics.

$$
\begin{aligned}
\vec{v}_{rr} &= v_x\hat{\imath} + v_y\hat{\jmath} + \omega\hat{k} \times (r_{rr_x}\hat{\imath} + r_{rr_y}\hat{\jmath}) \\
\vec{v}_{rr} &= v_x\hat{\imath} + v_y\hat{\jmath} + \omega r_{rr_x}\hat{\jmath} - \omega r_{rr_y}\hat{\imath} \\
\vec{v}_{rr} &= (v_x - \omega r_{rr_y})\hat{\imath} + (v_y + \omega r_{rr_x})\hat{\jmath}
\end{aligned}
$$

Project the rear-right wheel onto its wheel vector.

$$
\begin{aligned}
v_{rr} &= ((v_x - \omega r_{rr_y})\hat{\imath} + (v_y + \omega r_{rr_x})\hat{\jmath}) \cdot \frac{\hat{\imath} - \hat{\jmath}}{\sqrt{2}} \\
v_{rr} &= ((v_x - \omega r_{rr_y}) - (v_y + \omega r_{rr_x}))\frac{1}{\sqrt{2}} \\
v_{rr} &= (v_x - \omega r_{rr_y} - v_y - \omega r_{rr_x})\frac{1}{\sqrt{2}} \\
v_{rr} &= (v_x - v_y - \omega r_{rr_y} - \omega r_{rr_x})\frac{1}{\sqrt{2}} \\
v_{rr} &= (v_x - v_y - \omega(r_{rr_x} + r_{rr_y}))\frac{1}{\sqrt{2}} \\
v_{rr} &= \frac{1}{\sqrt{2}}v_x - \frac{1}{\sqrt{2}}v_y - \frac{1}{\sqrt{2}}\omega(r_{rr_x} + r_{rr_y})
\end{aligned} \tag{11.11}
$$

This gives the following inverse kinematic equations.

$$
\begin{aligned}
v_{fl} &= \frac{1}{\sqrt{2}}v_x - \frac{1}{\sqrt{2}}v_y - \frac{1}{\sqrt{2}}\omega(r_{fl_x} + r_{fl_y}) \\
v_{fr} &= \frac{1}{\sqrt{2}}v_x + \frac{1}{\sqrt{2}}v_y + \frac{1}{\sqrt{2}}\omega(r_{fr_x} - r_{fr_y}) \\
v_{rl} &= \frac{1}{\sqrt{2}}v_x + \frac{1}{\sqrt{2}}v_y + \frac{1}{\sqrt{2}}\omega(r_{rl_x} - r_{rl_y})
\end{aligned}
$$

$$v_{rr} = \frac{1}{\sqrt{2}}v_x - \frac{1}{\sqrt{2}}v_y - \frac{1}{\sqrt{2}}\omega(r_{rr_x} + r_{rr_y})$$

Now we'll factor them out into matrices.

$$\begin{bmatrix} v_{fl} \\ v_{fr} \\ v_{rl} \\ v_{rr} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}}(r_{fl_x} + r_{fl_y}) \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}(r_{fr_x} - r_{fr_y}) \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}(r_{rl_x} - r_{rl_y}) \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}}(r_{rr_x} + r_{rr_y}) \end{bmatrix} \begin{bmatrix} v_x \\ v_y \\ \omega \end{bmatrix}$$

$$\begin{bmatrix} v_{fl} \\ v_{fr} \\ v_{rl} \\ v_{rr} \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 & -(r_{fl_x} + r_{fl_y}) \\ 1 & 1 & (r_{fr_x} - r_{fr_y}) \\ 1 & 1 & (r_{rl_x} - r_{rl_y}) \\ 1 & -1 & -(r_{rr_x} + r_{rr_y}) \end{bmatrix} \begin{bmatrix} v_x \\ v_y \\ \omega \end{bmatrix} \tag{11.12}$$

11.6.2 Forward kinematics

Let $\mathbf{M}$ be the 4×3 inverse kinematics matrix above including the $\frac{1}{\sqrt{2}}$ factor. The forward kinematics are

$$\begin{bmatrix} v_x \\ v_y \\ \omega \end{bmatrix} = \mathbf{M}^+ \begin{bmatrix} v_{fl} \\ v_{fr} \\ v_{rl} \\ v_{rr} \end{bmatrix} \tag{11.13}$$

where $\mathbf{M}^+$ is the pseudoinverse of $\mathbf{M}$.

11.7 Swerve drive kinematics

A swerve drive has an arbitrary number of wheels which can rotate in place independent of the chassis. The forces they generate are shown in figure 11.4.

11.7.1 Inverse kinematics

$$\begin{aligned} \vec{v}_{wheel1} &= \vec{v}_{robot} + \vec{\omega}_{robot} \times \vec{r}_{robot2wheel1} \\ \vec{v}_{wheel2} &= \vec{v}_{robot} + \vec{\omega}_{robot} \times \vec{r}_{robot2wheel2} \\ \vec{v}_{wheel3} &= \vec{v}_{robot} + \vec{\omega}_{robot} \times \vec{r}_{robot2wheel3} \\ \vec{v}_{wheel4} &= \vec{v}_{robot} + \vec{\omega}_{robot} \times \vec{r}_{robot2wheel4} \end{aligned}$$

where $\vec{v}_{wheel}$ is the wheel velocity vector, $\vec{v}_{robot}$ is the robot velocity vector, $\vec{\omega}_{robot}$ is the robot angular velocity vector, $\vec{r}_{robot2wheel}$ is the displacement vector from the